

RobotX 2026: Breaking the Surface

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Abstract— Historically focused on subsea robotics, Team Mecatron marks its debut at RobotX 2026 by breaking the surface to expand across the full maritime operational envelope. We introduce our tri-domain System of Systems (SoS), composed of our Uncrewed Surface Vehicle (USV) *Poseidon*, Uncrewed Aerial Vehicle (UAV) *Phoenix*, and Uncrewed Underwater Vehicle (UUV) *Kraken*. Underpinning this heterogeneous architecture is our custom multi-protocol communications and networking architecture, engineered to ensure robust inter-agent synchronization and seamless cross-domain integration.

Recognizing that this as our inaugural campaign at RobotX, we center our development framework around ‘dynamic adaptability.’ Rather than assuming flawless execution, we systematically force system failures and induce edge cases during field trials to aggressively expose and learn from our operational vulnerabilities. This deliberate strategy of failing fast enables us to retire failure modes early, ultimately forging a highly resilient foundational baseline capable of rapid fault recovery on the water.

I. INTRODUCTION

Representing Nanyang Technological University (NTU) in Singapore at the 2026 Maritime RobotX Challenge, Team Mecatron engineered a unified, multi-domain System of Systems (SoS). To overcome the asymmetric operational challenges of cooperative maritime environments, we transitioned our foundational subsea expertise into a synchronized tripartite fleet. This distributed architecture ensures our surface, aerial, and underwater agents seamlessly execute complex, interdependent missions.



Fig. 1: UAV-Phoenix (left), USV-Poseidon (center), and UUV-Kraken (right)

II. COMPETITION STRATEGY

Our RobotX 2026 strategy extends our extensive multi-UUV experience into a coordinated fleet. This transition introduces more than additional operating domains – execution now depends on transferring observations, navigation objectives

The complete team roster, subteam breakdowns, and specific engineering contributions are detailed in Appendix B.

TABLE I: Planned task allocation and target capability

Task	Platforms	Lower Target Tier	Highest Target Tier	Scope
1: Safe Passage	USV, UAV	Core	Advanced	Primary
2: Survey and Repair	UUV, UAV	Core	Advanced	Primary
3: Coordinated Logistics	USV, UAV	Advanced	Disruptive	Primary
4: Incident Response	USV, UUV, UAV	Core	Core	Contingent

and mission outcomes between vehicles. Development concentrates on the cross-vehicle interfaces our selected tasks require, with platform-specific capability built upon shared software and electrical foundations.

A. Task Priorities and Capability Allocation

Tasks 1–3 form our primary development scope, targeting Advanced capabilities for Safe Passage and Survey and Repair, and Disruptive capabilities for Coordinated Logistics. Task 4 remains contingent on qualification status, vehicle readiness, and battery condition. Table I summarizes these commitments.

Task 1, Safe Passage: Phoenix detects buoys from an elevated viewpoint and relays target coordinates to Poseidon, which fuses them with onboard LiDAR perception and Real-Time Kinematic (RTK) Global Navigation Satellite Systems with Inertial Navigation Systems (INS) (RTK-INS) localization to navigate. Success depends on converting an aerial observation into a consistent and usable navigation input.

Task 2, Survey and Repair: Kraken performs underwater pipeline following using downward-facing perception and Doppler Velocity Log (DVL)-aided localization, engaging the damaged section with a magnetic probe while Phoenix awaits the resulting payload delivery request.

Task 3, Coordinated Logistics: Poseidon uses camera and LiDAR observations to identify target docking bays and guide water-shooter engagement, while Phoenix performs aerial delivery. Perception, positioning and payload actuation needs to operate in combination rather than in isolation.

Task 4, Incident Response: We retain this task as a conditional objective and will attempt it based on our vehicle availability, operating condition, and the competition situation. This approach limits any additional scope without presenting an unvalidated capability as a baseline.

B. Coordination and Mission Dependencies

Each vehicle executes its own instance of the autonomy stack on a separate Robot Operating System 2 (ROS 2) domain

to enable independent testing and fault containment. Poseidon orchestrates the fleet, coordinating delegated UAV and UUV missions via behavior-tree actions during multi-domain tasks. This architecture prioritizes direct information handoffs: the sensing vehicle translates observations into spatial references the receiving vehicle can immediately utilize. Furthermore, every delegated mission must return a definitive execution status upon completion.

C. Development Priorities and Design Trade-Offs

Our primary development priority is establishing reliable end-to-end execution across the entire fleet. While we strive to maximize physical field testing to validate sensor performance and vehicle dynamics, we must balance this against the strict time constraints of parallel hardware assembly. As a necessary trade-off, we offload initial integration dependencies—such as localization, navigation, and cross-vehicle mission logic—into our UnityMDS [1] simulation environment. This measured approach allows software development to advance rapidly without waiting for physical fleet completion.

Mechanically, while we strive to equip each vehicle with highly specialized task capabilities, we must balance these additions against the risk of escalating system complexity. Consequently, our priority is to confine custom fabrication strictly to physical interactions that standard navigation and perception cannot perform. We accept specific trade-offs where necessary: we tolerate the additional actuation and control overhead of the water shooter to gain aiming freedom beyond vessel heading, and we rely on passive compliance in the magnetic probe and UAV gripper to balance reliable payload handling against mechanical simplicity. To mitigate the risk of these custom additions, we prioritize reliability elsewhere by reusing proven electrical functions and structural arrangements across the fleet.

D. Readiness and Contingency Planning

Targeting an Advanced or Disruptive capability tier does not guarantee field deployment; operational readiness strictly requires independent validation of all underlying task dependencies. Our testing sequence progresses rigidly from isolated component checks to vehicle-level operations, culminating in coordinated multi-agent missions. Aerially supported navigation clears validation only when the observation-to-navigation data handoff proves seamless, while physical interactions demand successfully synchronized target acquisition, positioning, and actuation.

To preserve fleet resilience, the system architecture isolates faults at both vehicle and mission levels. A localized hardware failure may disable a specific platform and its dependent tasks, but the surviving fleet remains fully operational. Ultimately, field execution decisions dynamically weigh any lost capabilities against remaining task dependencies, strictly prohibiting unvalidated platform substitution.

III. DESIGN STRATEGY

Operating across three distinct domains introduces additional sensing, control, and coordination requirements. Our

fleet therefore couples vehicle-specific systems with standardized hardware architectures and shared autonomy frameworks, thereby enhancing overall system reliability, operational robustness, and ease of field deployment.

A. Mechanical Design

We combine reusable structural architectures with task-specific interaction mechanisms to prioritize maintenance access and rapid field reconfiguration. To limit system complexity, we strictly confine custom fabrication to physical interactions that standard navigation and perception cannot execute.

1) *Shared Architecture & Adaptations*: For instance, Poseidon and Kraken share a planar thruster configuration, with four thrusters positioned at the corners of a square layout and canted at a nominal 30° (Figure 2). Reusing this geometry leverages our prior RoboSub development [2] and establishes a common framework for control allocation, though discrepancies in mass distribution, hydrodynamics, and operating domains still necessitate platform-specific tuning.

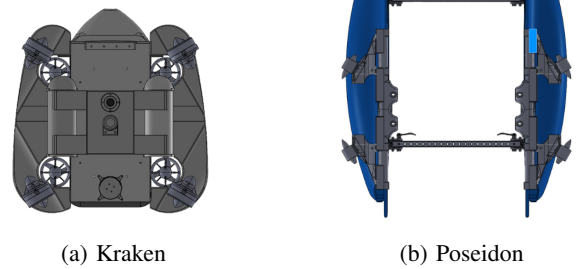
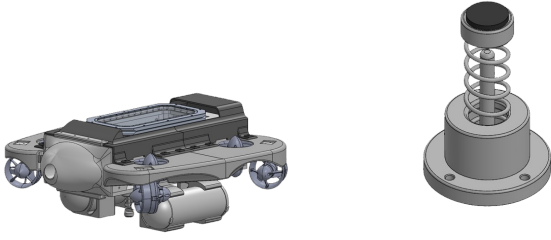


Fig. 2: Bottom views of Kraken and Poseidon

2) *UUV RobotX Mechanical Adaptations*: To adapt the Kraken platform from our RoboSub 2026 baseline for RobotX, we executed several targeted mechanical modifications. We replaced the original chassis with a rigid CNC-machined frame (Figure 3a) to secure the updated payload architecture. Transitioning to the new mission parameters, we removed the legacy torpedo bays and integrated a passively compliant magnetic probe (Figure 3b). Because actuating underwater switches requires sustained contact against surfaces the UUV may approach off-axis, the probe utilizes a central linear guide and compression spring to absorb impact and prevent lateral buckling. An elastic rubber boot press-fits the magnet to the assembly, permitting tool-less field swaps to match competition switch sensitivity. Furthermore, to satisfy multi-domain coordination and safety mandates, we expanded the surface communications mast to house four dedicated antennas (Bullet, LTE, E-stop, and Teleoperation) and mounted a high-visibility, all-round status light above the waterline to continuously broadcast the vehicle's operational state.

3) *Active USV Gimbal*: Forcing a USV to constantly reposition a fixed nozzle for minor aiming corrections severely degrades station-keeping stability. To resolve this, we integrated an active 2-Degree-Of-Freedom (DOF) gimbal onto Poseidon to steer the water jet (Figure 4a). A compact pump delivers

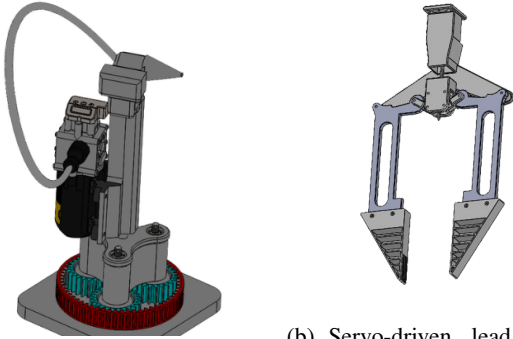


(a) Kraken's modular mechanical assembly (b) Spring-loaded magnetic probe

Fig. 3: Kraken UUV mechanical assembly and payload actuator

pressurized flow to the nozzle, while a planetary gearbox actuates base azimuth and pitch angle independently. This decoupled mechanism allows the vessel to hold its heading and position while precisely directing pressurized flow onto the target.

4) *UAV Airframe & Payload Actuation*: We constructed Phoenix's airframe using carbon-fiber spars and low-infill 3D-printed brackets to shed structural weight without sacrificing avionics rigidity. For payload retrieval, a single servo-driven lead screw actuates symmetric gripper jaws (Figure 4b).



(a) USV's water shooter

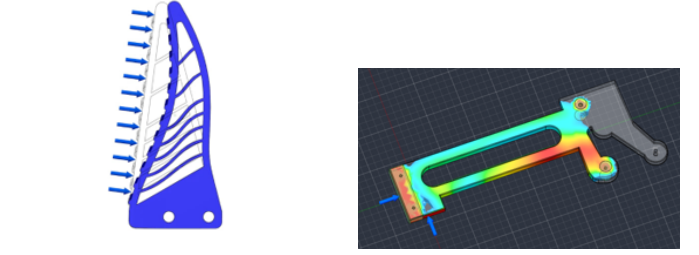
(b) Servo-driven, lead screw-actuated gripper

Fig. 4: Payload actuation mechanisms for the USV and UAV

Because aerial positioning over a target inherently drifts, we integrated compliant TPU fingers based on the Fin Ray effect [3] to passively deform around the payload and absorb off-center contact errors (Figure 5a). To further minimize aerial mass, we applied topology optimization to the gripper's structural arms (Figure 5b), successfully shedding weight while preserving critical load paths around all pivot and linkage connections.

B. Electrical & Embedded Systems

1) *Power Architecture*: To isolate sensitive computing payloads from inductive motor noise and brownout transients, we bifurcate every electrical system we design into two dedicated domains: the Compute Electrical Subsystem (CESS), which



(a) Expected deformation of the TPU Fin Ray finger (b) Topology optimization of the gripper arm

Fig. 5: Structural analysis and optimization of Phoenix gripper components

powers sensors and compute nodes, and the Actuation Electrical Subsystem (AESS), which drives high-current actuators. To streamline hardware development, maintenance, and field deployment, we developed two modular power architectures (Figure 6) that we standardize across all our operational vehicles, auxiliary ground units, and supporting electrical infrastructure.

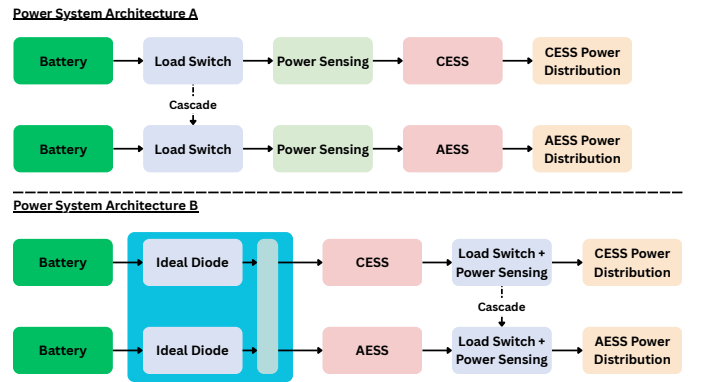


Fig. 6: Power Architectures A and B

Every system onboard Poseidon, Phoenix, Kraken, our Operator Control Station (OCS) and, Mobile Command Centres (MCCs) integrates multiple layers of electrical protection. Each battery pack carries a Battery Management Systems (BMSs) with integrated software and hardware Over Current Protection (OCP) to cut current during over-current events, Over Voltage Protection (OVP) and Under Voltage Lock Out Protection (UVLO) to prevent over-charge and over-discharge.

Through our experience with building UUVs, we have learnt that albeit sharing similar up and downstream architectures, AESS and CESS systems have inherently different requirements: AESS requires short-duration high-current ($\approx 40A$) bursts while CESS requires constant low-current ($\approx 8A$) draws. This observation pushed Mecatron to utilize different battery chemistries and configurations for different utilities.

To enhance the endurance of Poseidon, "Power System Architecture A" is chosen. Equipped with a 6S 12500mAh Solid State battery for CESS, we effectively double our CESS

capacity and thus endurance compared to utilizing our pre-existing 4S 6200mAh LiPo battery.

To support continuous flight on Phoenix even after rigorous high-current draw maneuvers, the AESS onboard Phoenix bypasses OCP, OVP and UVLO protections, while keeping OCP for CESS. Hence, Phoenix adopts "Power System Management B" with slight modifications: it takes power from a single 6S battery and connects to CESS without Ideal Diode and AESS without Load Switch. Phoenix's electrical system can be studied in subsection B-C.

As Kraken has two battery cylinders on its underside, to maintain buoyancy and balance, we opted to utilize the same battery solutions in all our UUV battery cylinders. Thus, to maximize our endurance as well handle hot-swapping of batteries, we opted for 'Power System Architecture B' in Kraken. This allows for quicker maintenance turnover as well as deployment times.

2) *Safety & Teleoperation Systems*: To satisfy stringent safety requirements for RobotX, we developed a modular, plug-and-play wireless Emergency Stop module that operates in tandem with the physical manual Emergency Stop switches installed on our vehicles. We enforce a fail-safe architecture where the default steady state is inherently safe: AESS power remains cut unless the onboard Microcontroller (MCU) continuously receives valid heartbeat packets from our handheld transmitter. Consequently, triggering either the physical switch or the wireless cutoff, losing signal, encountering a subsystem fault, or disconnecting battery power immediately depowers all thrusters and actuators. We detail the complete safety architecture in subsection B-C.

To achieve independent manual intervention across operating domains, we adapted the custom EduBoat [4] remote controller developed by Mechatron. This modified handheld controller allows operators to seamlessly take over and teleoperate either Poseidon or Kraken during launch, recovery, and complex near-field maneuvers.

3) *Vehicle-Specific Integration*: We adapt the common electrical architecture to each vehicle to satisfy domain-specific spatial, thermal, and mechanical constraints.

To meet strict payload and space limitations in Phoenix, we prioritized mass reduction and thermal dissipation across all circuit designs. On its Power Distribution Unit (PDU), we integrated solid copper busbars rather than wide PCB traces to carry high continuous currents while minimizing board footprint and localized heat buildup.

To navigate tight volumetric limits within Kraken's cylindrical aluminum hull, we omitted bulky busbars and instead engineered an ultra-compact, high-density PDU footprint. We integrated this PDU into a custom backplane with modular slide-in trays, eliminating cable clutter, enabling rapid bench servicing, and embedding dedicated optical leak-detection sensors to trigger immediate safety shutdowns upon water ingress.

We distributed Poseidon's electronics between the central electronics box and the two pontoon hulls, isolating high-power distribution from sensitive compute nodes and facilitat-

ing parallel subsystem testing. For vehicle locomotion, we are currently experimenting with closed-loop current control using Vedder Electronic Speed Controllers (VESC) [5] to regulate thruster torque directly, providing more precise station-keeping and dynamic trajectory tracking. Furthermore, to ensure consistent nozzle pressure and repeatable trajectory profiles during target-blasting tasks, we engineered a 2-DOF gimbal driven by a custom closed-loop current-controlled driver that directly governs pump torque.

C. Software & Autonomy

Our RoboSub 2026 software stack was developed for a single class of vehicle, the UUV. RobotX 2026 requires the same autonomy framework to operate across three heterogeneous physical domains and to support coordination between them. We extend the software architecture to preserve a common autonomy interface while isolating platform-specific differences in sensing, localization, and communication.

1) *Unified Multi-Domain Architecture*: The autonomy stack must support surface, underwater, and aerial platforms without distinct implementations of planning and perception. Although the three vehicles differ in their controllable axes and available sensors, the underlying autonomy logic remains common. We encode domain-specific behaviours through plugin implementations and parameter selection rather than platform-specific code paths.

In RobotX 2026, we extend the use of Navigation3 (Nav3), our current suite of 3D planners, controllers, and custom behaviours implemented as Navigation2 (Nav2) [6] plugins, to all domains. Previously deployed only on UUVs, Nav3 will operate on multiple vehicle domains for the first time at RobotX 2026, without changes to the plugin interfaces. The UUV and UAV use 3D planners and controllers (Figure 7), while the USV utilises the 2D counterparts and enables the LiDAR-based costmaps required for obstacle avoidance.

We apply the same abstraction to perception. Object distance is estimated from depth measurements within YOLO11 [7] instance masks, while the depth source is provided through a platform-specific plugin. The USV projects LiDAR returns into the camera image using the live camera-LiDAR extrinsic transform (Figure 8); the UUV and UAV use monocular metric depth from Depth Anything 3 [8]. Downstream object processing remains identical despite the heterogeneous sensing hardware.

2) *Fleet-Consistent Localization*: Inter-vehicle information exchange requires all platforms to express their state and observations within a common spatial reference. Hence, we localize all vehicles with respect to a common map frame and select the underlying state estimator according to the operating domain. The USV and UAV use dual-antenna RTK-GNSS to achieve centimetre-level position and sub-degree heading. The UUV, for which GNSS is unavailable, retains the SR-UKF [9] developed for RoboSub that fuses IMU delta-velocity, DVL velocity, and pressure depth to generate an external odometry estimate for the autopilot. This common frame convention removes the need for vehicle-specific coordinate conversion

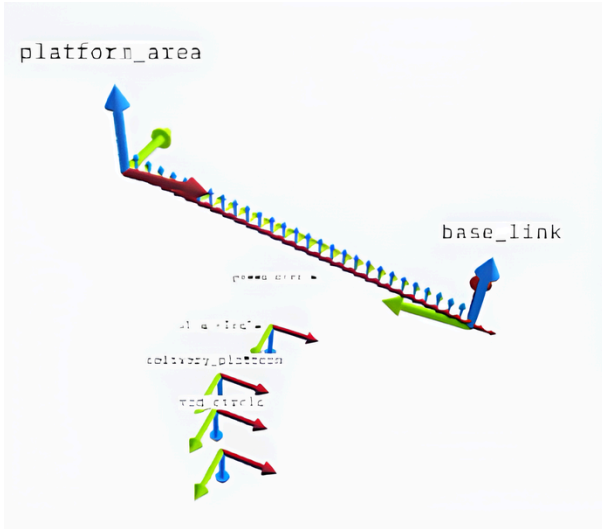


Fig. 7: UAV (base_link) 3D spatial path planning and following using Nav3

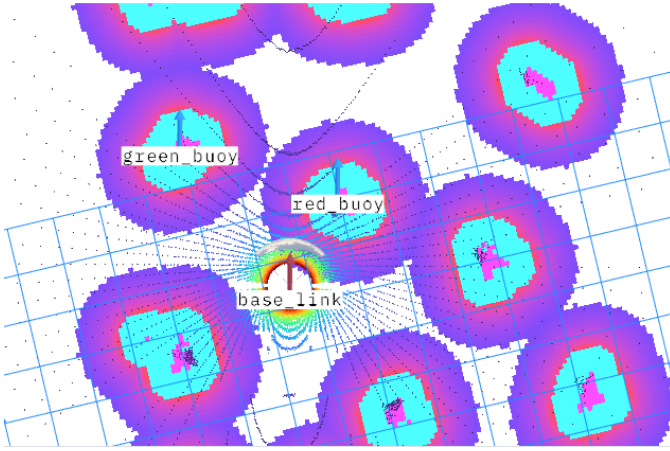


Fig. 8: USV LiDAR-based navigation: obstacle observations in the local costmap

during task execution: a buoy detected by the UAV during Safe Passage can be represented directly as a navigation goal by the USV.

3) *Cross-Vehicle Mission Delegation*: Vehicle coordination should extend from the Behavior Tree [10] model that already governs single-vehicle autonomy, rather than introduce a separate mechanism for distributed execution. We expose each vehicle's Behavior Tree executor as a ROS 2 action server that can be invoked across vehicle domains. A remote mission is represented as a single action node within the USV master Behavior Tree, preserving the same sequencing, timeout, retry, and cancellation semantics used for local behaviours. We bridge observations from other vehicles as coordinate frames into the receiving vehicle's transform tree, and register the remotely detected target in the navigation stack through the same interface as a local observation. During subsea tasks, the UUV retains mission observations until task completion. The

UUV will then resurface and attempt to relay the information to the other vehicle.

D. Fleet Communication

Fault isolation between platforms is a prerequisite for coordinated operation. At RoboSub, both vehicles shared a single ROS 2 domain; each vehicle now occupies its own, with only whitelisted topics, services and actions bridged between them. This restricts cross-link traffic to the information coordination requires, and confines a saturated or failed link to a single platform.

1) *Physical Communication Architecture*: The physical network links the OCS directly to all three vehicles and RoboCommand through a central Ethernet switch. The OCS houses dedicated operator consoles for each vehicle, as well as an on-board computer that interfaces directly with RoboCommand. Wireless communication between the OCS and the fleet is established via a shore-side WiFi access point communicating with WiFi stations mounted on each vehicle.

2) *Modularity and Scalability*: To facilitate independent subteam testing and prevent network saturation, we structured the fleet's communications around a modular architecture. Each vehicle operates its own self-contained local network, explicitly bridging only mission-critical information among each other. Because this architecture is decentralized, any platform can dynamically connect to or disconnect from the broader system without destabilizing the communication links of the remaining vehicles. This segmented approach effectively scales our prior single-vehicle networking setup into a highly resilient, distributed SoS spanning air, surface, and subsea domains.

IV. TESTING STRATEGY

Mecatron's testing strategy combines simulation-based autonomy development, component and subsystem checks, and integrated field testing. This approach allows software development to proceed alongside hardware fabrication, while separating component-level issues from vehicle integration and cross-vehicle coordination. The planned sequence progresses from standalone checks to vehicle-specific testing and finally to full-system operation.

A. Simulation and Virtual Testing

1) *Multi-Vehicle Simulation Environment*: Validating a surface, underwater and aerial fleet requires a simulation environment combining vehicle dynamics, sensor generation and flight-controller behaviour. No existing simulator covers all three domains with autopilot integration. Hence, Mecatron extended UnityMDS [1], our in-house multi-drone, multi-domain maritime simulator, to support the complete RobotX fleet. Each vehicle runs its own ArduPilot Software-in-the-Loop (SITL) instance, reproducing the autopilot firmware and its sensor fusion. Unity generates camera, LiDAR, GNSS, IMU and DVL data, and models drag and added mass from the hull mesh. The simulator is bridged to ROS 2 so that the

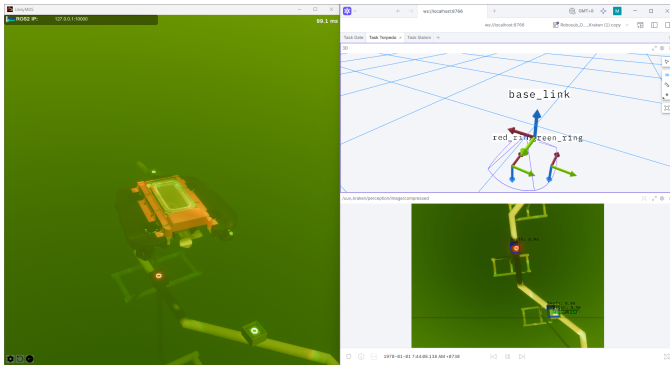


Fig. 9: UUV operating in a simulated murky underwater environment in UnityMDS

autonomy stack operates against simulated sensors. Murky-water and bright-light scenarios (Figure 9) extend testing to the environmental conditions expected at the competition.

2) *Autonomy Development and Mission Rehearsal*: Water and air time is the scarcest resource in the schedule. Therefore, a separate simulated implementation of the autonomy logic would introduce divergence with real deployment. The autonomy code above the drivers is shared between simulation and deployment, with a single configuration flag selecting the operating environment. Full cross-vehicle missions can be rehearsed before committing vehicle time, and the shared configuration provides a basis for comparing behavior between simulation and physical deployment.

B. Component and Subsystem Testing

Board-level faults become difficult to distinguish from wiring, packaging or subsystem-interaction problems once a vehicle is assembled. The electrical testing outline follows a staged sequence: each Printed Circuit Board (PCB) is fabricated and tested independently, then tested under the constraints of its intended vehicle, and finally tested as part of the integrated system.

We rigorously test standalone PCB prior to vehicle integration. Each PDU was tested up to its rated current limit, with voltage drop, over-voltage threshold, temperature rise and efficiency recorded throughout. Each procedure is tied to the board's intended function and to documented acceptance criteria. The Safety System PCB was tested with spare components to confirm that the fail-safe mechanism operated as intended.

The second planned stage evaluates components within the electrical and mechanical constraints of each vehicle. Boards were assembled with their designed mounts to establish sub-team cohesion per vehicle, and physical layouts were modified accordingly. The final stage combines the subsystems to assess their operation together.

Software verification follows the same principle. Every deployment is preceded by an automated component-level test suite implemented using ROS 2 `launch_testing`. The suite is shared across the fleet and parameterized per

vehicle platform, with individual tests verifying physical and functional properties:

- **Sensors**: the IMU is verified to report the expected gravity magnitude and a level orientation while stationary, while all sensor streams are checked for valid and up-to-date timestamps. Vehicle-specific checks cover the DVL on the USV and UUV, GNSS and LiDAR on the USV, and GNSS on the UAV.
- **Actuation**: each thruster or motor is individually commanded through ArduPilot's motor test functionality to verify correct actuation.
- **Localization**: on the UUV, the state estimate is validated against prescribed forward, lateral, and vertical motions to verify the consistency of the localization system.

This procedure isolates faults at the component level before vehicle deployment, rather than during integrated field operations.

C. Full System and Field Testing

Field testing assesses whether behaviour developed in simulation transfers to the physical vehicles, and whether the integrated fleet can execute coordinated missions. Localization field checks precede subsequent testing, providing a basis for assessing the state estimates on which navigation and mission execution depend. The field-testing record should connect observed behavior to design decisions. Where a test exposes a limitation, the report should identify the condition that produced it, the resulting software or hardware modification, and the outcome of the subsequent retest.

V. CONCLUSION

Team Mecatron's road to the 2026 Maritime RobotX Challenge has been an incredible test of adaptability and teamwork. Facing a tight turnaround immediately after RoboSub, we successfully built and integrated a collaborative fleet comprising our USV, UAV, and UUV.

Ultimately, Poseidon, Phoenix, and Kraken operate not as a collection of independent platforms, but as a cohesive, decentralized network. By deliberately stress-testing this architecture to retire failure modes early, we have established a highly adaptable, robust operational baseline fully prepared to execute complex, cooperative missions in unstructured maritime environments.

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REFERENCES

- [1] T. L. Vu, M. C. Togamulia, and S. D. Upadhyay, “Unitymds: a high-fidelity multi-drone, multi-domain simulator for maritime robotics,” in *OCEANS 2026 Sanya*, 2026, pp. 1–10.
- [2] NTU Mecatron, “Robosub 2026: Pushing the frontier,” Nanyang Technological University, Tech. Rep., 2026.
- [3] W. Crooks, G. Vukasin, M. O’Sullivan, W. Messner, and C. Rogers, “Fin ray® effect inspired soft robotic gripper: From the robosoft grand challenge toward optimization,” *Frontiers in Robotics and AI*, vol. 3, 2016.
- [4] S. J. X. Cheng, Z. Low, Timothy, Nicole, Joowan, Yingzhan, and Yongzhan, “Mecatron EduBoat: An Open Source Boat.” [Online]. Available: <https://github.com/NTU-Mecatron/EduBoat-Workshop>
- [5] B. Vedder, “VESC – Open Source ESC,” <http://vedder.se/2015/01/vesc-open-source-esc/>, 2015.
- [6] S. Macenski, F. Martín, R. White, and J. Ginés Clavero, “The Marathon 2: A navigation system,” in *Proc. IEEE/RSJ IROS*, 2020.
- [7] G. Jocher, J. Qiu, and A. Chaurasia, “Ultralytics YOLO,” 2023.
- [8] H. Lin *et al.*, “Depth Anything 3: Recovering the visual space from any views,” 2025, arXiv:2511.10647.
- [9] R. van der Merwe and E. A. Wan, “The square-root unscented Kalman filter for state and parameter-estimation,” in *Proc. IEEE ICASSP*, 2001.
- [10] M. Colledanchise and P. Ögren, *Behavior Trees in Robotics and AI: An Introduction*. CRC Press, 2018.
- [11] H. W. Kuhn, “The Hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.
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APPENDIX B

TEAM ROSTER AND CONTRIBUTIONS

A. Team Leads

Scott Jia Xu Cheng — Executive Head, Head of Digital Software
 Do Phuoc Truong Nguyen — Executive Head
 Yi Liang Soon — Head of Mechanical
 Gerardo Nandavardhana Abdipranoto — Head of Mechanical
 Cheng Jing Hao — Head of Electrical and Embedded Systems
 Hai Duong Trinh — Head of Electrical and Embedded Systems
 Chuanhua Liu — Head of Robotics Software
 Gia Minh Nhan — Head of Robotics Software
 Zi Xuan Tan — Head of Business Operations, Head of Digital Software
 Poh Hian Chee — Head of Logistics and Finances

B. Team Advisors

Gia Bao Bui — Robotics Software Team Advisor
 Wei Ming Wong — Robotics Software Team Advisor
 Tien Luc Vu — Robotics Software Team Advisor

C. Team Members

Electrical and Embedded Systems: Jerry Ruoxu Zhang, Hoang Quan Pham, Aaron Wei Ren Ong, Tyla Tran, Brian Iskandar

Robotics Software: Hari Sarvajana, Jia Qian Peh, Hong Son Nguyen, Zheng Khai Hong, Moolya Baba Purandara, Daniel Wang, Le Anh Duc Vu, Moolya Sai Purandara, Moshes Yosepa Manihuruk, Jia Hui Lim, Gregory Hans Nugraha, Sung Ho Park, Teik Fei Chan, Tuan Minh Nguyen

APPENDIX A
SPONSORS

A. Gold Sponsors

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B. Silver Sponsors

- DSO National Laboratories — For their generous monetary support.
- VectorNav — For their generous equipment sponsorship.
- WaterLinked — For their generous equipment support.

C. Bronze Sponsors

- Zen4Blue — For their constant support and trust in Mecatron’s mission.

Digital Software: Zhuo Li Eu, Yun Tai Quek, Jun Jie Soh, Bryan Xu Hen Ng, Min Han Delvin Ho, Xin Ying Yeong, Devansh Poddar, Shantanu Agrawal, Poorneshwar Karthikeyan, Sanvi Agarwal, Tang Yao Goh

Mechanical: Ambrose Wye Yang Chan, Mahendran Jeevan, Kanda Shotaro, Soen William Brantley, Sanjay Sharavanan, Yi Hang Ooi, Logesh Muresh, Zexuan Zhang , Zi Jun Leong , Shaun Jia Xuan Tee, Wei Hang Tan , James Jun Yi Lee, Nicholas Yu Xuan Tee, Ian Nikos Gunawan, Marvel Christian Kusuma, Gabriel Ren Jie Woon, Jian Xian Lee , Dimitry Marvello, Swayanshu Patra, Khade Shantanu, Yuetao Chong, Zhe Yu Yik , Wen Jet Yap , Shirley Bee Yun Gan, Jing En Wong, Nur Jannah Aisyah Medina Romisah K H Torbett, Kuan Fan, Jia En Lee, Nathan Gabriel Winoto, Jeremy Winata Suhendra, Heinrich Tanuwijaya, Julian Kok Tian Leow, Joanna Stevia Chandrasaputra

APPENDIX C ELECTRICAL TESTS PLAN

Components	Checklist	Pass/Fail
Load Switch	Power-On: Apply battery, enable ENPWR Power-Off: Disable ENPWR Fuse Continuous Rating: Load output to 25A Fuse Trip Point: Ramp current above 25A Reverse Current Block: Apply reverse voltage at output, switch OFF TVS Clamping: Apply transient pulse at input TVS Clamping: Apply transient pulse at input Status LED: Toggle power state	Output = input voltage within delay Output drops to 0V within delay No trip, temperature stays rated Trips per time-current curve No significant reverse current Clamps near rated voltage, board survives Clamps near rated voltage, board survives LED matches state
Ideal-Diode-OR	Single-Battery Power-On: Connect only battery input 1 Single-Battery Power-On: Connect only battery input 2 Dual-Battery Function: Connect both at different voltages Hot-Swap Transient: Disconnect one battery under load while other remains connected Reverse Current Blocking (each channel): With one battery disconnected, check for backfeed into its input	Output = battery 1 voltage Output = battery 2 voltage Output follows the higher-voltage battery Output voltage stays in regulation, no glitch/reset downstream No significant reverse current
Protection System	ELRS Module Power: Connect ELRS receiver to its connector, measure its power pin UART Link (MCU and ELRS): Power up, monitor ELRS TX/RX activity Wireless E-Stop Command Response: Send command from RC transmitter Wireless Failsafe: Power off/Disconnect RC transmitter	5V present Valid UART both directions Wireless e-stop signal toggles to correct logic level for each command Wireless e-stop signal defaults to STOP-safe

TABLE II: Finite Element Analysis For Components.

Category	Testing Steps
Communication	1) Pinging PDUs, Actuator, Acoustics 2) Check if there is connection
Physical E-Stop and PDU	- Physical E-Stop Switch Continuity - PDU Logic Truth Table: Cycle all combinations of physical switch and wireless e-stop states, measure PDU-facing output

TABLE III: Integration Testing Features.

APPENDIX D

USV ELECTRICAL ARCHITECTURE BLOCK DIAGRAMS

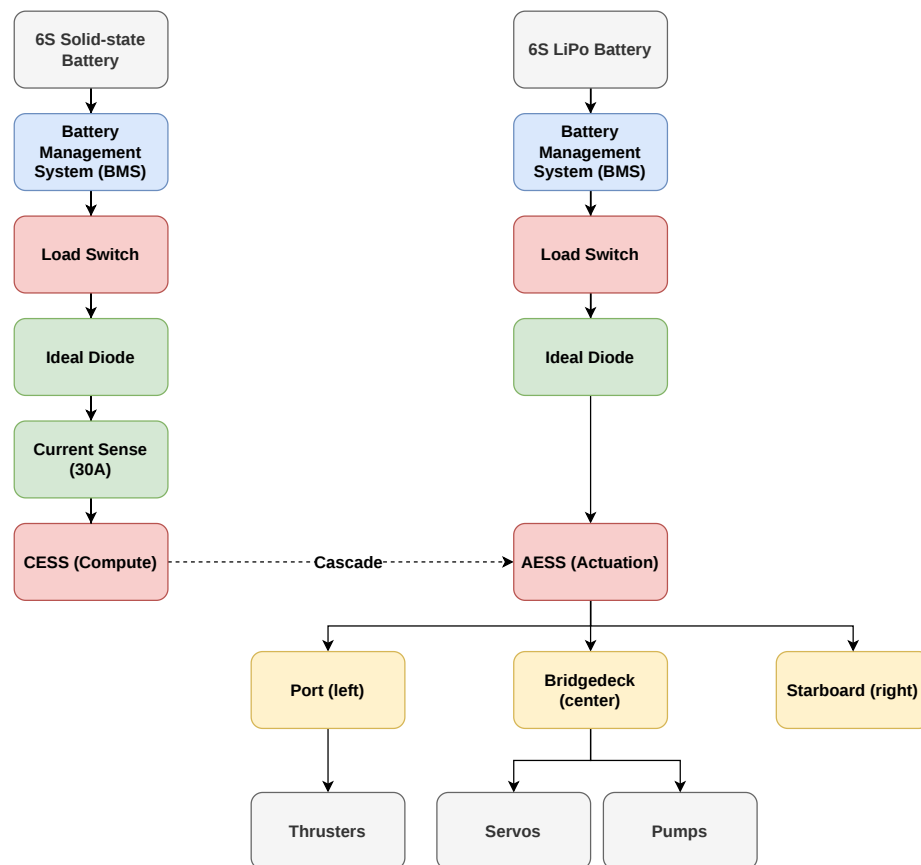
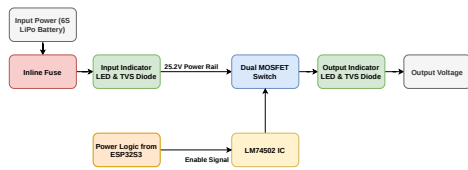
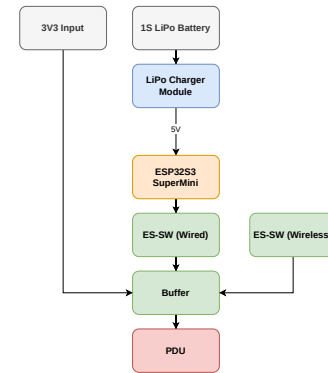


Fig. 10: USV Electrical System Architecture

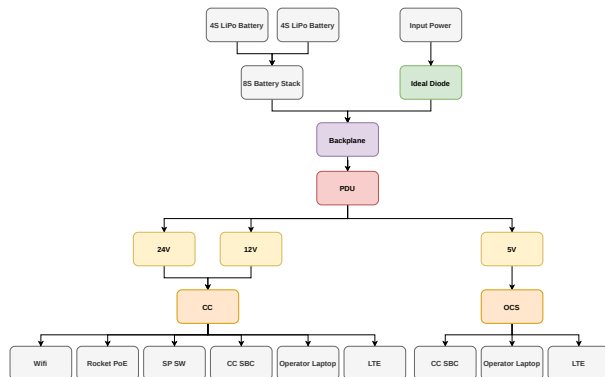
- **PDU:** The main function of PDU is to take the main 6S battery power source and distribute it to the AESS and the CESS. For all three vehicles, their AESS and CESS are similar in concept, with the AESS delivering high-current power to the



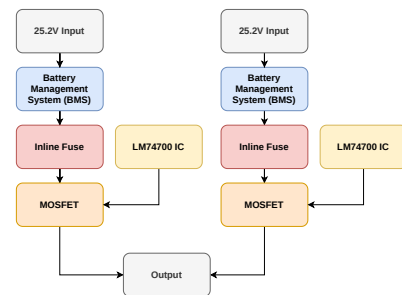
(a) Load Switch Architecture



(b) Wireless Emergency Stop Architecture



(c) Uninterruptible Power Supply Architecture



(d) Ideal Diode OR Circuit Architecture

Fig. 11: USV Electrical Subsystem Architectures

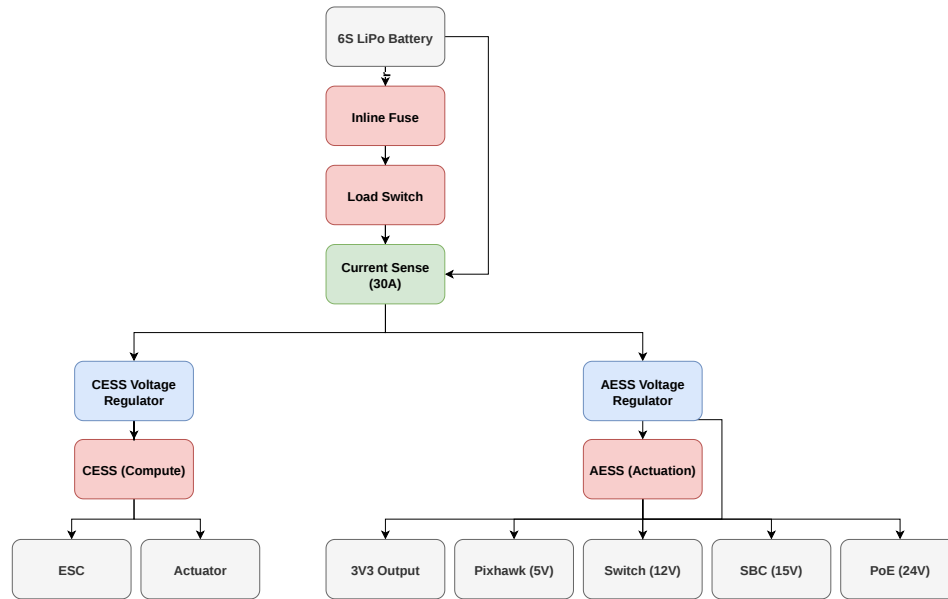


Fig. 12: UAV Power Distribution Unit Architecture

thrusters and actuators; while the CESS delivers stable and clean voltages to our compute units and sensors, such as the Pixhawk, Single Board Computer (SBC), Power over Ethernet (PoE) and physical switch.

- Power-Path-OR:** The Power-Path-OR PCB lies within the current path, between the BMS and PDU. In terms of IO, the board hosts two XT90 male inputs, and an XT90 male output, as well as a vertical JST-XH two-pin for a multimeter plug. The board routes current in from the two XT90 inputs, each connected to a 6S battery with 25.2V at full charge. These inputs are passed through 80A fuses, and then each through a MOSFET wired with an IC to act as ideal diodes. This circuit ensures that the batteries do not back charge each other when at slightly different voltages. The outputs are routed together, and there are decoupling capacitors on both inputs and both outputs. This common output is routed through a shunt resistor, with Kelvin connections to an INA226. This IC connects via I2C to an ESP32-S3 on the board, which interprets current flow and transmits wirelessly off the board. 5V is provided with an XL-1509 buck converter, and then to 3V3 using the ESP32's internal buck. 25V2 inputs, outputs and 5V have LED indicators. All relevant components are specified around 60A continuous, 100A spike.
- Load Switch:** The Load Switch is a compact, high-current power management PCB created for 6S LiPo battery systems (25.2V max) that serves as the primary power gateway for all vehicle subsystems. The input power passes directly through an inline fuse for overcurrent safety before reaching the LM74502 controller. By driving dual back-to-back MOSFETs, the controller provides true bidirectional isolation, preventing reverse current back-feeding and completely disconnecting downstream electronics when powered off. To maintain continuous system reliability, the board implemented robust transient and signal protection features such as TVS diodes to clamp high-voltage inductive spikes. Small LEDs were also implemented as indicators to see whether the PCB is working.
- Protection System:** The Protection System consists of an on-board Kill Switch and a Remote Kill Switch. Both systems operate in a failsafe manner, which means that if the E-Stop switch is activated, any parts of the system fail, the battery fails, or is removed, the vehicle's actuators will be immediately disconnected from power. To achieve this, we utilise a buffer with Enable Pin and design a handheld remote control that continuously sends heartbeats to the on-board MCU. The Protection PCB outputs LOW to the Load Switch PCB when the Safety System is activated.

APPENDIX E
3D MODELS OF ELECTRONIC BOARDS

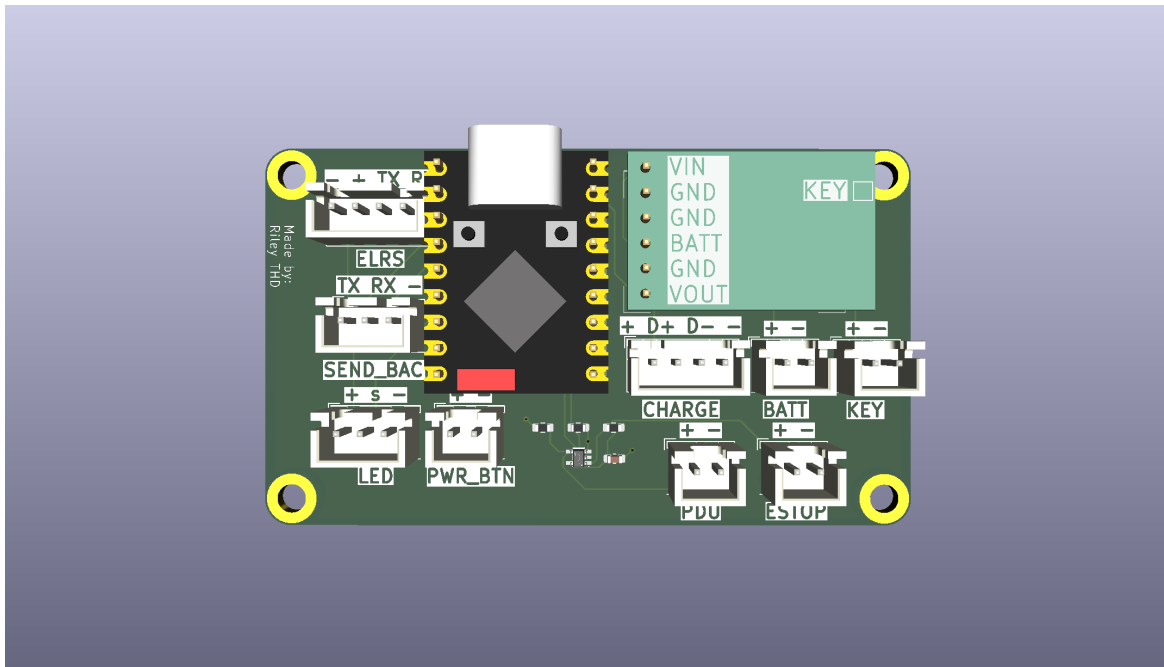


Fig. 13: Protection System Board

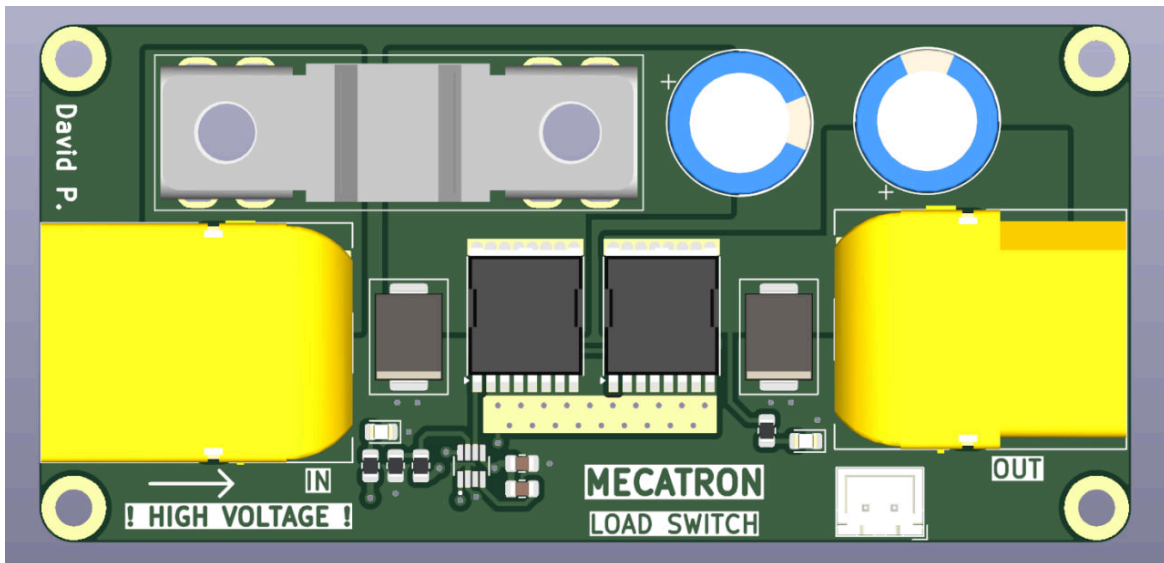


Fig. 14: Load Switch